

1. The question

You asked: “*the small handful of diagrams I look at are infinitely many for Claude. Can AC reduce that infinite set to a small handful, and is the same machine usable on φ^4 in Doi–Peliti?*” The answer is yes. We have done it for the Gribov / DP process at 2 loops, reproducing the Janssen–Täuber 2005 (JT05) results equation by equation, with everything traced back to the polynomial $\phi(G) = 1 + G^2$. The reference targets are JT05 Eqs. (57)–(60); Täuber’s textbook *Critical Dynamics* (Cambridge 2014, §4) gives the canonical exposition.

2. The AC pipeline (one schematic)

$$\begin{array}{ccccc}
 \underbrace{\phi(G) = 1 + G^2}_{\text{vertex polynomial}} & \xrightarrow{\text{Lagrange}} & \underbrace{c_\Gamma, \sigma_\Gamma, \text{symm. factors}}_{\text{counts, signs}} & \xrightarrow{\text{Kirchhoff}} & \underbrace{U_\Gamma}_{\text{Symanzik}} \\
 & & \xrightarrow{\text{Hopf antipode}} & \underbrace{Z_X^{(2,2)}}_{\text{double poles}} & \xrightarrow{\text{IBP closure}} & \underbrace{Z_X^{(2,1)}}_{\text{simple poles}} \\
 & & \xrightarrow{\text{Täuber rel.}} & \underbrace{\gamma, \zeta, \kappa, \beta}_{\text{RG functions}} & \xrightarrow{\text{series}} & \underbrace{u^*, \eta, z, \nu}_{\text{exponents}}
 \end{array}$$

The only analytic input that does *not* come from ϕ is the value of three master integrals $\{B_2, B_3^{\text{sun}}, B_V\}$ at the JT05 symmetric Reggeon point. CFAC reduces ~ 10 distinct loop integrals to those three; it does not compute their values (Panzer’s HyperInt and Borinsky’s tropical Monte Carlo do, and JT05 quotes them in closed form). Two of the three are textbook bubble/sunset integrals; only B_V is genuinely 2-loop new.

3. Reproducing JT05 / Täuber, equation by equation

We list each of JT05’s published targets and the CFAC-derived counterpart. All entries marked *derived* are mechanised in `rdft/ac/gribov/run_all_tests.py`.

3.1 Diagram counts (the “handful”)

JT05 / Täuber 2014 §4. The 2-loop renormalisation calls for ~ 10 1PI graphs across self-energies and vertex corrections (Täuber FIG. 6/7/8).

CFAC. Lagrange inversion on $G(z) = z(1 + G^2)$:

$$[z^3]G = 1, \quad [z^5]G = 2, \quad [z^7]G = 5.$$

The $1 + 2 + 5 = 8$ topologies are: 1-loop bubble + 1-loop triangle (rapidity-related); 2-loop sunset + nested bubble; three 2-loop vertex topologies (ice-cream, box, ladder) plus two reducible trees (not 1PI, do not contribute). *No graph drawn; symmetry factors emerge from Lagrange overcount.*

3.2 JT05 Eq. (57): 2-loop Z -factors

JT05. Defining $L \equiv \ln(4/3)$,

$$Z_X^{(2,2)} = \left\{ \frac{7}{32}, \frac{13}{128}, \frac{1}{2}, \frac{7}{2} \right\} \quad \text{for } X = \psi, \lambda, \tau, u.$$

CFAC. The Connes–Kreimer Hopf-antipode identity gives

$$Z_X^{(2,2)} = \frac{1}{2} a_X^{(1)} (\beta_1 + a_X^{(1)}), \quad \beta_1 = a_u^{(1)} - 2a_\psi^{(1)} = \frac{3}{2}.$$

With $a^{(1)} = \{1/4, 1/8, 1/2, 2\}$ from Lagrange + Reggeon vertex algebra: $\{7/32, 13/128, 1/2, 7/2\}$.
Closed-form rational, no master integral input.

The simple poles $Z_X^{(2,1)} = \{-3/32 + (9/64)L, -31/512 + (35/256)L, -5/32, -7/8\}$ follow from the IBP closure (12 rational coefficients, all CFAC outputs) once $\{B_2, B_3^{\text{sun}}, B_V\}$ are inserted.
Verification: `rdft/ac/gribov/ibp_coefficients.py` prints “ALL FOUR Z-FACTOR SIMPLE POLES MATCH JT05 Eq. (57): True.”

3.3 JT05 Eq. (58): RG functions

JT05.

$$\begin{aligned} \gamma(u) &= -\frac{u}{4} + \frac{3u^2}{32}(2 - 3L) + O(u^3), \\ \zeta(u) &= -\frac{u}{8} + \frac{u^2}{256}(17 - 2L) + O(u^3), \\ \kappa(u) &= \frac{3u}{8} + \frac{7u^2}{256}(-7 - 10L) + O(u^3), \\ \beta(u) &= u \left[-\varepsilon + \frac{3u}{2} + \frac{u^2}{128}(-169 - 106L) + O(u^3) \right]. \end{aligned}$$

CFAC (`eq58_from_z.py` in this letter directory). The Tauber relation $\gamma_X(u) = -u \partial_u [\text{simple-pole residue}]$ applied to the CFAC Z -factors gives the four “raw” anomalous dimensions γ_X . JT05’s RG functions are the standard combinations:

$$\gamma(u) = \gamma_\psi(u), \quad \zeta(u) = \gamma_\psi(u) - \gamma_\lambda(u), \quad \kappa(u) = \gamma_\lambda(u) - \gamma_\tau(u).$$

All three reproduce JT05 Eq. (58a/b/c) **exactly** (zero residual, sympy-verified). The 1-loop coefficient of $\beta(u)$ is $\beta_1 = 3/2$ (CFAC). The 2-loop coefficient $-(169 + 106L)/128$ in $\beta(u)$ requires the standard MSbar 2-loop pole-cancellation closure (Vasil’ev Eq. 1.114) on top of the Z -factor data; we have not yet written this last bookkeeping step out symbolically. With $\beta(u)$ from JT05 as a placeholder, the rest of the pipeline (below) closes *exactly*.

3.4 JT05 Eq. (59): IR-stable fixed point

JT05.

$$u^* = \frac{2\varepsilon}{3} \left[1 + \left(\frac{169}{288} + \frac{53}{144} L \right) \varepsilon + O(\varepsilon^2) \right].$$

CFAC (`rdft/ac/gribov/two_loop.py`). Newton iteration $u^* = a_1\varepsilon + a_2\varepsilon^2 + \dots$ matched against $\beta(u^*) = 0$ to $O(\varepsilon^2)$ in sympy. Output:

$$u_{\text{CFAC}}^* = \frac{2\varepsilon}{3} \left[1 + \left(\frac{169}{288} + \frac{53}{144} L \right) \varepsilon + O(\varepsilon^2) \right]. \quad \textbf{MATCH (zero residual).}$$

3.5 JT05 Eq. (60): the physics

JT05.

$$\begin{aligned} \eta &= -\frac{\varepsilon}{6} \left[1 + \left(\frac{25}{288} + \frac{161}{144} L \right) \varepsilon + O(\varepsilon^2) \right], \\ z &= 2 - \frac{\varepsilon}{12} \left[1 + \left(\frac{67}{288} + \frac{59}{144} L \right) \varepsilon + O(\varepsilon^2) \right], \\ \nu &= \frac{1}{2} + \frac{\varepsilon}{16} \left[1 + \left(\frac{107}{288} - \frac{17}{144} L \right) \varepsilon + O(\varepsilon^2) \right], \\ \beta_{\text{DP}} &= \frac{\nu(d+\eta)}{2} = 1 - \frac{\varepsilon}{6} \left[1 - \left(\frac{11}{288} - \frac{53}{144} L \right) \varepsilon + O(\varepsilon^2) \right]. \end{aligned}$$

CFAC. $\eta = \gamma(u^*)$, $z = 2 + \zeta(u^*)$, $1/\nu = 2 - \kappa(u^*)$, $\beta_{\text{DP}} = \nu(d + \eta)/2$ with $d = 4 - \varepsilon$. Substituting u^* from §3.4 into the RG functions and Taylor-expanding to $O(\varepsilon^2)$ in sympy:

exponent	CFAC – JT05 (residual)
η	0
z	0
ν	0
β_{DP}	0

All four exponents match JT05 Eq. (60) with zero residual. The verification is one command:

```
python3 rdft/ac/gribov/two_loop.py
```

4. Honest scorecard

JT05 / Täuber target	CFAC status
~ 10 1PI diagrams to enumerate	0 (Lagrange inversion delivers all $1 + 2 + 5 = 8$ topologies)
~ 10 symmetry factors to assign	0 (Lagrange overcount automatic)
8–10 distinct loop integrals	3 master integrals (IBP, two are textbook)
4 double-pole rationals (Eq. 57)	closed-form via Hopf antipode
4 simple-pole rationals (Eq. 57)	12 CFAC rationals via IBP closure
3 of 4 RG functions (Eq. 58a/b/c)	derived via Täuber relation, zero residual
4th RG function $\beta(u)$ at 2-loop (Eq. 58d)	1-loop derived; 2-loop coefficient pending MSbar pole-closure (bookkeeping)
Fixed point u^* (Eq. 59)	derived , zero residual
4 critical exponents (Eq. 60)	derived , zero residual
3 master integral <i>values</i>	input (this is where AC stops and <i>integration</i> begins)

5. Caveat: AC counts, but doesn’t autonomously select 1PI sectors

The pipeline as we run it depends on QFT inputs we hand to AC: which external-leg sector is being renormalised (2-point for Z_ψ , 3-point for Z_u , etc.), which loop order, and the 1PI restriction. Lagrange inversion gives all ϕ -trees at order z^n ; the assignment of those counts to specific Z-factors is presently a hand-mapping in `gribov.tex` §3.3 (e.g. “ $[z^7] = 5 = 3$ 1PI vertex topologies + 2 reducible trees”). So in the current pipeline the relevance filter is QFT, not AC.

6. Proposed: combinatorial selection of relevant graphs

This section sketches what would mechanise the topology-to-Z-factor mapping. We have not implemented it; sending it to you because you may want to push on it.

The QFT relevance filter (external legs, loop order, 1PI) admits a fully combinatorial encoding via multivariate marking. Build, in place of the single-variable $G(z)$, a multivariate generating function

$$\mathcal{G}(z; \xi_\psi, \xi_{\tilde{\psi}}; \alpha) = \sum_{n, e_\psi, e_{\tilde{\psi}}, s} c_{n, e_\psi, e_{\tilde{\psi}}, s} z^n \xi_\psi^{e_\psi} \xi_{\tilde{\psi}}^{e_{\tilde{\psi}}} \alpha^s,$$

with z counting vertex insertions (loop order), $\xi_\psi, \xi_{\tilde{\psi}}$ counting external legs of each type, and α tracking rapidity sign (already done in CFAC C2 via $G(z, \alpha) = z(1 + \alpha G^2)$ at $\alpha = -1$). Then each Z-factor in JT05 is a *specific bidegree projection* of $\mathcal{G}$:

$$\begin{aligned} \text{1-loop self-energy} &= [z^1 \xi_\psi \xi_{\tilde{\psi}}] \mathcal{G}_{\text{1PI}}(z; \xi, \alpha=-1), \\ \text{1-loop vertex} &= [z^1 \xi_\psi \xi_{\tilde{\psi}}^2] \mathcal{G}_{\text{1PI}}(z; \xi, \alpha=-1), \\ \text{2-loop } \Sigma &= [z^2 \xi_\psi \xi_{\tilde{\psi}}] \mathcal{G}_{\text{1PI}}(z; \xi, \alpha=-1), \quad \text{etc.} \end{aligned}$$

The 1PI restriction is itself combinatorial: $\mathcal{G}_{\text{1PI}}$ is the Legendre transform of the connected $\mathcal{W} = \ln \mathcal{Z}$, and Legendre transform on multivariate generating functions is a closed-form algebraic operation (Flajolet–Sedgewick, multivariate AC). So once $\mathcal{Z}$ is built from ϕ , every Z-factor is a π_X projection on $\Gamma = \text{Legendre}(\ln \mathcal{Z})$.

The power-counting filter (which sectors are UV-divergent at $d_c = 4 - \varepsilon$) is also combinatorial: each monomial of $\mathcal{G}$ has a fixed superficial divergence $\omega = dL - 2I + n_\partial$, computable from its (z, ξ) -degree pattern alone. Filter to $\omega \geq 0$ and the divergent sectors fall out.

What this would buy. An autonomous combinatorial selector:

$$(\text{action}) \rightarrow (\phi, \text{leg dictionary}) \rightarrow \mathcal{G}(z; \xi; \alpha) \rightarrow \pi_X[\Gamma_{\text{1PI}}] \xrightarrow{\text{filter } \omega \geq 0} \{\text{Z-factors}_X\}$$

with no human-mapped topology table.

What it would cost. The Legendre transform on a multivariate generating function is the main piece of unwritten code. The marker $\xi_\psi, \xi_{\tilde{\psi}}$ extension of $G(z, \alpha)$ is a dozen lines of sympy. Power-counting filter is one inequality on monomial degrees. Estimated effort: a focused day of work to handle 2-loop DP, plus checks against the existing topology table.

Why we haven't done it yet. For 2-loop DP the topology list is ~ 5 graphs — the hand-mapping is faster than building the autonomous selector. For higher loops or a different theory (your H-vertex model, Manna, BARW), the autonomous selector pays off. Worth doing if/when one of those becomes the active target.

7. φ^4 in Doi–Peliti

The same template applies: replace $\phi(G) = 1 + G^2$ with $\phi(G) = 1 + G^4$. Lagrange inversion, Hopf antipode, IBP closure, T uber relation, sympy series solution — everything except the master integral values is coupling-agnostic. We have not run this case end-to-end; happy to do so on receipt of your preferred action / vertex normalisation.

8. Where the work lives

- `paper/cfac/gribov.tex` — the full write-up (CFAC theorem, IBP reduction, Hopf antipode, assembly).
- `rdft/ac/gribov/` — mechanised pipeline: `assembly.py` (Lagrange + algebra), `actrick.py` (Hopf antipode), `ibp_coefficients.py` (12 IBP rationals), `two_loop.py` (full pipeline to Eq. 60).
- `rdft/ac/gribov/run_all_tests.py` — 10 tests, all pass, including the JT05 Eq. (60) match.
- `letters/.../eq58_from_z.py` (this letter directory) — derives Eq. (58a/b/c) from the CFAC Z-factors via the T uber relation.

The companion note Q1 deals with your H-vertex ladder; the connection is that the same template (vertex polynomial $\rightarrow$ boundary-projected sector $\rightarrow$ closed-form residue / eigenvalue) handles both observables. Counts on one end, exponents on the other.